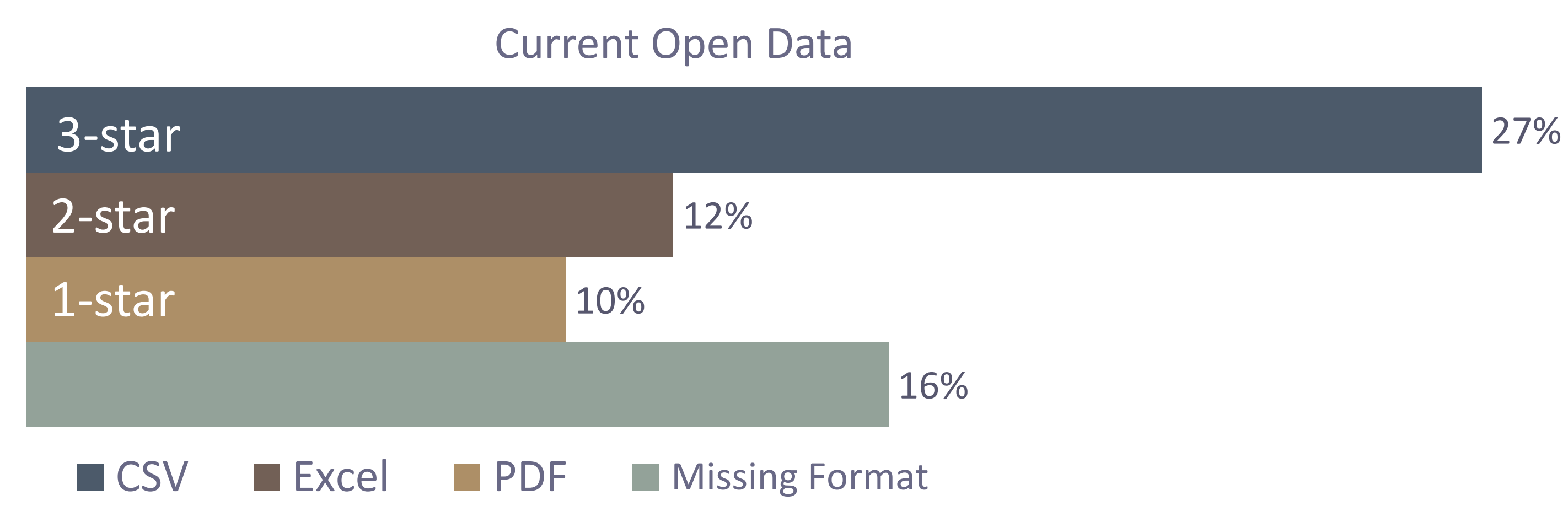


INTRODUCTION

Open Data has become a driver for publicly available data on the Web. More and more data – from governments, public institutions but also from the private sector – is made available online and is mainly published in so called Open Data portals. Yet, most of the data published on Open Data portals cannot be considered as Linked (Open) Data [1]:



Current Open Data is mainly published as 1- to 3-star data: openly licensed data, available in machine-readable and non-proprietary formats (according to Berner-Lee’s 5-star rating [2]). This fact motivates a further investigation of the potential of Semantic Web technologies to integrate and interlink this data.

PROBLEM STATEMENT

- Use of existing Semantic Web technologies to (*partially*) improve existing 3-star Open Data to 5-star data.
- Focus on tabular data: currently predominant in OD portals.

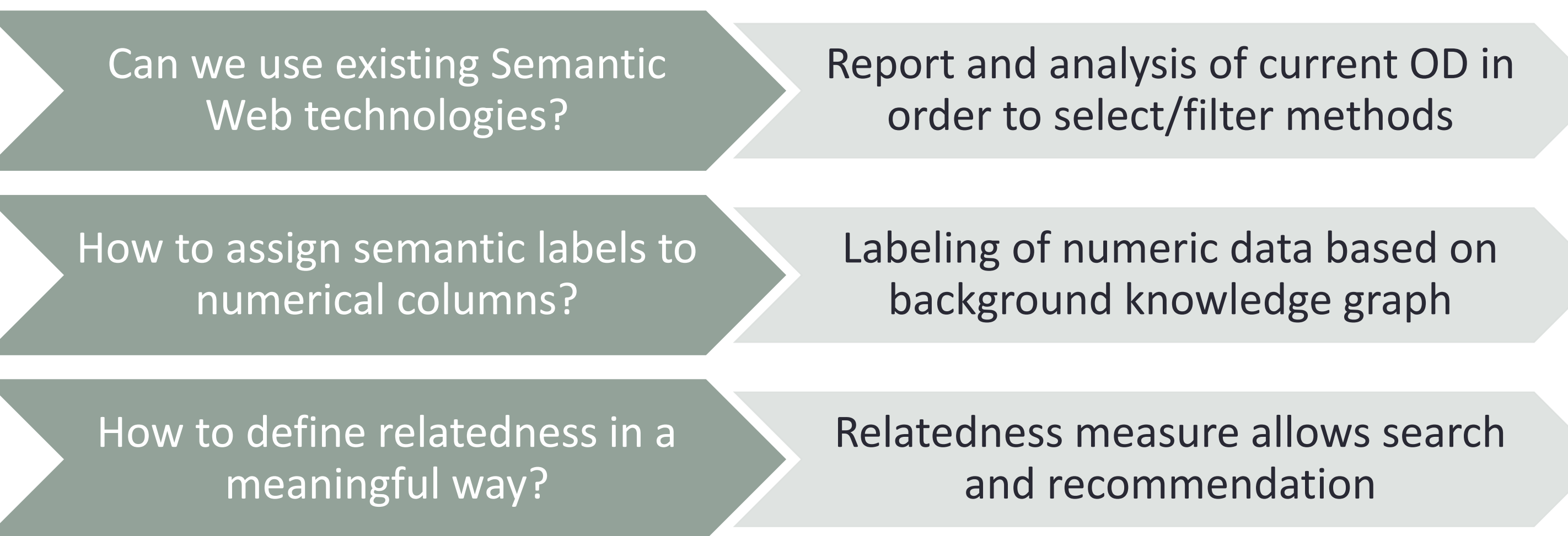
Men's Singles Rankings					
RK	PLAYER	COUNTRY	MOVEMENT	POINTS	
1	Novak Djokovic		↔	0	16950
2	Andy Murray		↔	0	8915
3	Roger Federer	Wohnungen in Wien, 2011			
4	Rafael Nadal	Dwellings in Vienna, 2011			
5	Stan Wawrinka	NUTS1	NUTS2	NUTS3	DISTRICT_CODE WHG_TOTAL
6	Kei Nishikori	AT1	AT13	AT130	90101 3004
7	Milos Raonic	AT1	AT13	AT130	90102 1049
8	Dominic Thiem	AT1	AT13	AT130	90103 1389
Table 1: Example from [3].		AT1	AT13	...	...

Table 2: CSV table on data.gv.at.

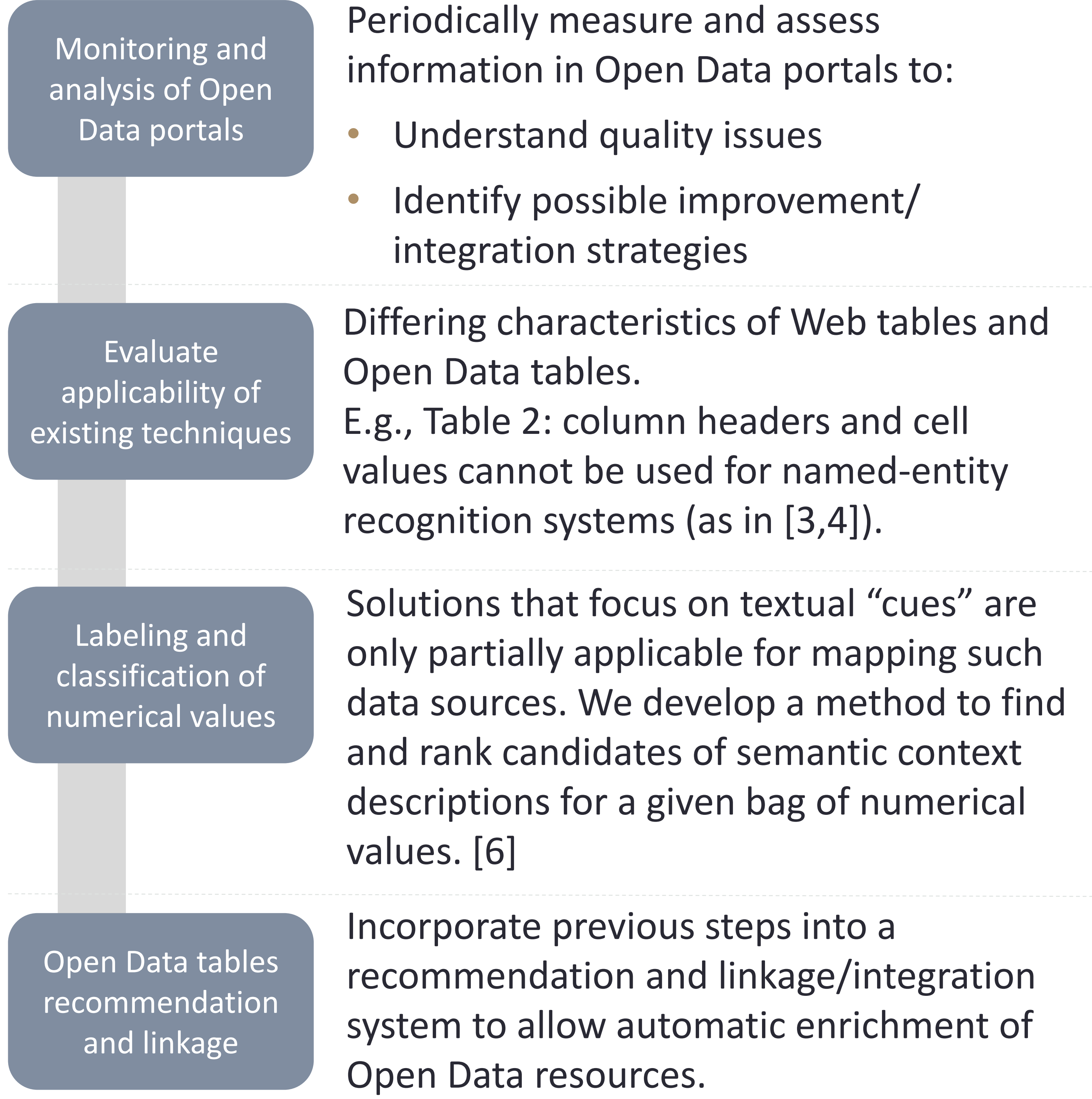
Most of the existing related work use Web/HTML tables as their corpus (e.g., Table 1) [3,4]. These approaches link column headers or cell values to properties/classes in ontologies or knowledge bases. However, typical in Open Data tables such textual descriptions (column headers or cell labels) are missing or cannot be mapped straightforwardly, particularly when they contain many numerical columns (e.g., see Table 2).

RESEARCH QUESTIONS & HYPOTHESES

The semantic analysis of Open Data CSVs increases the usability of Open Data and ideally generates 5-star linked data.



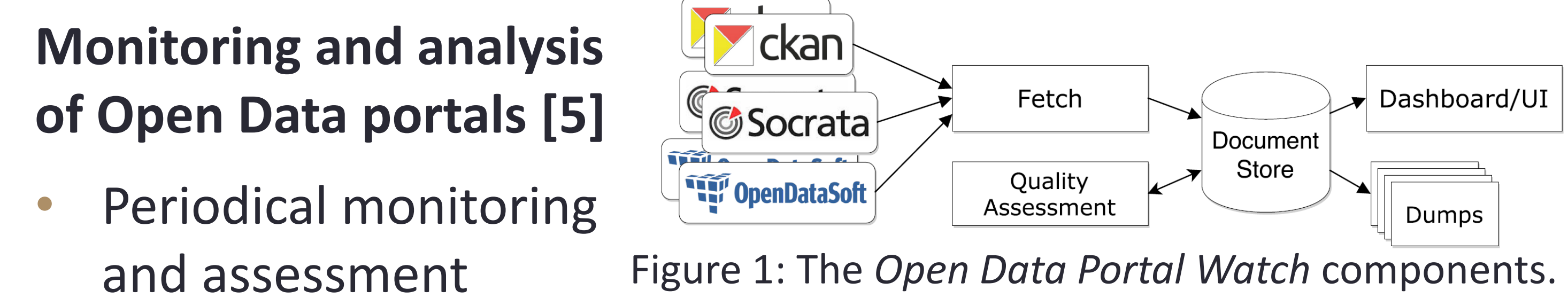
APPROACH



This work intends to *significantly* increase the usability of Open Data by *partially* enriching and relating tabular resources:

“a little semantics goes a long way” – James A. Hendler

PRELIMINARY RESULTS



- Periodical monitoring and assessment
- (Meta)data quality metrics, e.g.: retrievability, existence of contact or license information.
- <http://data.wu.ac.at/portalwatch/>

Labeling of numerical values [6]

- Assign multi-level semantic labels to numerical values
- Based on knowledge graph constructed from DBpedia

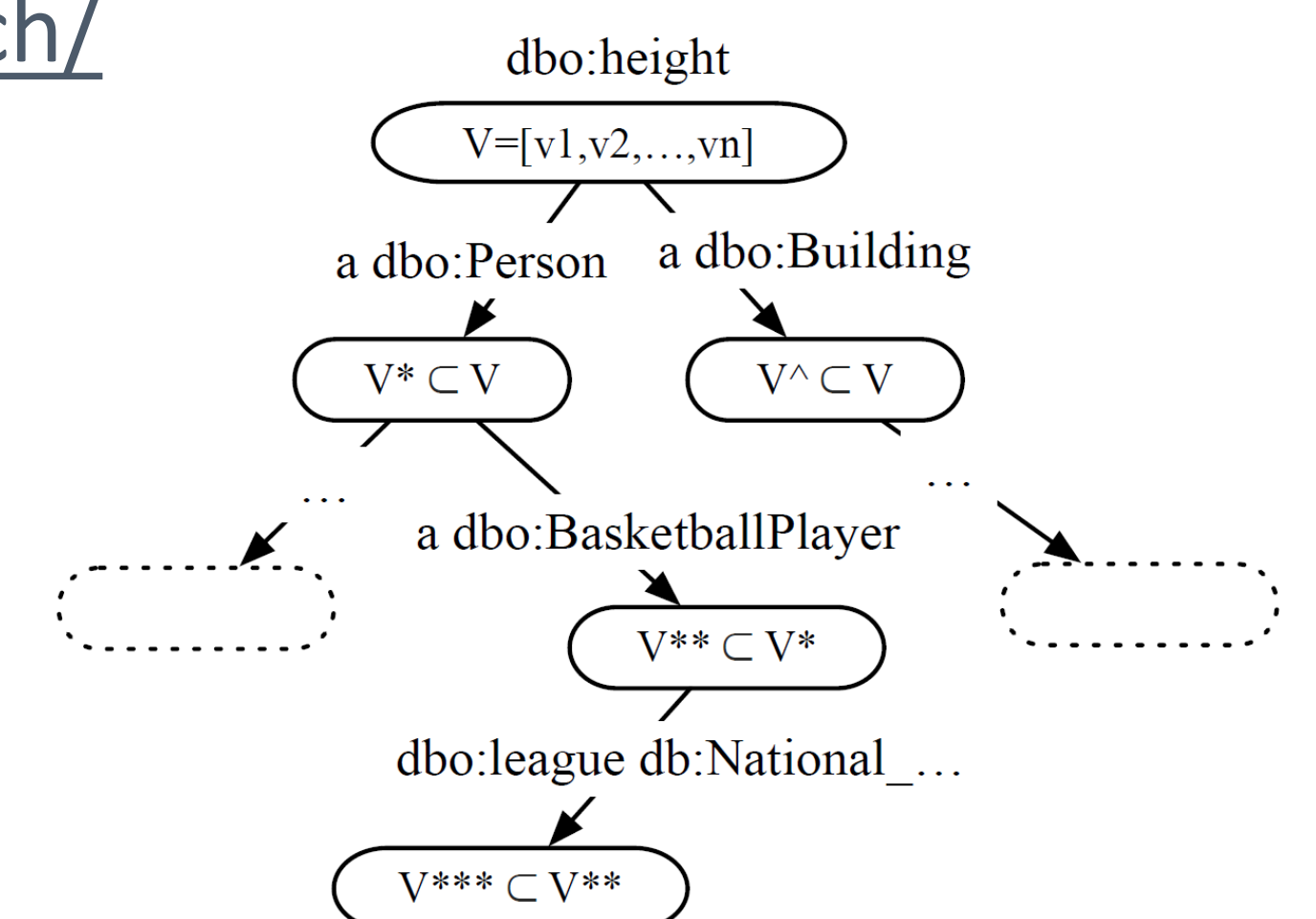


Figure 2: Hierarchical background knowledge.

REFERENCES

[1] Umbrich, J., Neumaier, S., Polleres, A.: Quality assessment & evolution of open data portals. In: The International Conference on Open and Big Data. IEEE, Rome, Italy (2015)

[2] Berners-Lee, T.: Linked data. <http://www.w3.org/DesignIssues/LinkedData.html> (2006)

[3] Das Sarma, A., et al.: Finding related tables. In: Proceedings of the 2012 ACM SIGMOD International Conference on Management of Data. pp. 817–828. ACM (2012)

[4] Zhang, Z.: Start small, build complete: Effective and efficient semantic table interpretation using tableminer. Semantic Web Journal (2014)

[5] Neumaier, S., Umbrich, J., Polleres, A.: Automated Quality Assessment of Metadata across Open Data Portals. ACM J. Data Inform. Quality (2016), to appear

[6] Neumaier, S., Umbrich, J., Parreira, J.X., Polleres, A.: Multi-level semantic labelling of numerical values. To appear in: International Semantic Web Conference (2016)